What is Claimed is:

1. A reusable cartridge for detecting an analyte in a solution, wherein the cartridge comprises:

a) a matrix,

b) detection molecules attached to the matrix, wherein the detection molecules specifically bind to the analyte, and wherein the detection molecules are optionally cross-linked to the matrix; and

c) at least one cryoprotectant,

wherein the cartridge can be used at least 10, 20, 50, or 100 times for detection of the analyte in a solution, and wherein, after each use the matrix is optionally stored with cryoprotectant in a dry state.

2. The reusable cartridge of claim 1, wherein the cartridge is a tube open at both ends, a tube open at one end, a well, a plate with or without wells, or a chip capable of containing the matrix.

3. The reusable cartridge of claim 1 or 2, wherein the matrix comprises particles (e.g. beads, grains, chips, or pellets) comprising silica or agarose, wherein the particles are optionally magnetic.

4. The reusable cartridge of any one of claims 1-3, wherein the matrix comprises protein A, protein G, and/or protein L.

5. The reusable cartridge of any one of claims 1-4, wherein the detection molecules are antibodies.

6. The reusable cartridge of any one of claims 1-5, wherein the detection molecules are cross-linked to the matrix.

7. The reusable cartridge of claim 6, wherein the detection molecules are cross-linked to the matrix with a crosslinker selected from selected from dimethylpimelimidate (DMP), a cyanate ester, an NHS ester, azlactone, carbonyl diimidazole (CDI), maleimide, iodoacetyl, pyridyl disulfide, hydrazide, or a carbodiimide.

8. The reusable cartridge of any one of claims 1-7, wherein the analyte detectable by the detection molecules on the matrix is a protein or peptide.

9. The reusable cartridge of claim 8, wherein the analyte is a major histocompatibility complex I (MHC-I) molecule.

10. The reusable cartridge of any one of claims 1-9, wherein the cartridge can be used at least 10 times for detection of a protein analyte in a cell lysate or biological fluid.

11. The reusable cartridge of any one of claims 1-10, which is stored in the dry state.

12. The reusable cartridge of claim 11, wherein the cartridge can be used at least 10 times for detection of a protein analyte in a cell lysate or biological fluid after storage in the dry state after each use.

13. The reusable cartridge of claim 11, wherein the cartridge can be used at least 20 times for detection of a protein analyte in a cell lysate or biological fluid after storage in the dry state after each use.

14. The reusable cartridge of claim 11, wherein the cartridge can be used at least 50 times for detection of a protein analyte in a cell lysate or biological fluid after storage in the dry state after each use.

15. The reusable cartridge of claim 11, wherein the cartridge can be used at least 100 times for detection of a protein analyte in a cell lysate or biological fluid after storage in the dry state after each use.

16. The reusable cartridge of any one of claims 11-15, wherein the matrix is stored at acidic pH.

17. The reusable cartridge of any one of claims 11-16, wherein the matrix is stored at 2-8 °C.

18. The reusable cartridge of any one of claims 1-10, wherein the matrix is stored in a wet state.

19. The reusable cartridge of claim 18, wherein the matrix is stored at acidic pH.

20. The reusable cartridge of claim 18 or 19, wherein the matrix is stored at 2-8 °C.

21. The reusable cartridge of any one of claims 1-20, wherein the cryoprotectant comprises one or more of sucrose, trehalose, ethylene glycol, propylene glycol, glycerol, 2-methyl-2,4-pentanediol (MPD), or dimethyl sulfoxide (DMSO).

22. A method of preparing a previously used cartridge according to any one of claims 1-21 for re-use, comprising, after analyte has been eluted from the cartridge, washing the matrix in 1-10% acetic acid, 1% formic acid, or 1% trifluoroacetic acid (TFA), washing the matrix in a buffer comprising cryoprotectant, and allowing the matrix to dry.

23. The method of claim 22, wherein the cryoprotectant comprises one or more of sucrose, trehalose, ethylene glycol, propylene glycol, glycerol, 2-methyl-2,4-pentanediol (MPD), or dimethyl sulfoxide (DMSO).

24. The method of claim 22 or 23, wherein the matrix is allowed to dry by heating the cartridge to at least 30 ℃, followed by storage at room temperature for at least 1 hour.

25. The method of claim 24, wherein the matrix is allowed to dry by heating the cartridge to 37 ℃, followed by storage at room temperature for at least 1 hour.

26. The method of any one of claims 22-25, wherein, after the matrix is allowed to dry, the cartridge is stored in a dry state at 2-8 ℃ until re-use.

27. A method of preparing a reusable cartridge according to any one of claims 1-21, the method comprising obtaining a cartridge comprising a matrix, contacting the matrix with the detection molecules, and crosslinking the detection molecules to the matrix.

28. A method of detecting an analyte in a solution, comprising contacting the matrix of the reusable cartridge according to any one of claims 1-21 with a solution comprising the analyte, and optionally eluting the analyte from the cartridge.

29. The method of claim 28, wherein the analyte is a peptide or protein.

30. The method of claim 29, wherein the analyte is an MHC-I molecule.

31. The method of any one of claims 28-30, wherein the solution is a biological fluid or cell lysate.

32. The method of claim 31, wherein the solution is filtered or treated to remove cell debris and membranous materials prior to contact with the matrix.

33. The method of any one of claims 28-30, wherein the cartridge has previously been used to detect the analyte at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 50, or 100 times prior to its use in the method.

34. The method of any one of claims 28-33, wherein, following elution of analyte, the cartridge is treated by the method of any one of claims 22-26.

35. A kit comprising at least one reusable cartridge according to any one of claims 1-21.

36. The kit of claim 35, further comprising: (a) at least one buffer; (b) at least one control cartridge without detection molecule or with a control detection molecule; (c) reagents for preparing the cartridge for storage and reuse; and/or (d) instructions for use.